

Xinyu Cheng

Ph.D. in Mathematics | Applied & Computational PDEs

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Education

Doctor of Philosophy in Mathematics <i>University of British Columbia</i> Area: Analysis, dynamics and numerics in PDEs. PhD Thesis: Analytical and numerical results for phase field, implicit free boundary, and fluid models. Advisers: Prof. Dong Li & Prof. Brian Wetton.	2021.11
Master of Science in Mathematics <i>University of British Columbia</i> Area: Analysis, dynamics and numerics in PDEs. MSc Thesis: On the Stability of a Semi-Implicit Scheme of Cahn-Hilliard Type Equations. Advisers: Prof. Dong Li & Prof. Brian Wetton.	2017.09
Bachelor of Science in Mathematics <i>The Chinese University of Hong Kong</i> Areas: Computational & Applied Mathematics; Enrichment Stream in Mathematics.	2015.11

Working Experience

Young Principal Investigator <i>Research Institute of Intelligent Complex Systems, Fudan University</i>	2023.12–present
Postdoctoral Fellow <i>School of Mathematical Sciences, Fudan University</i> Host: Prof. Zhen Lei.	2021.09–2023.11
Full-time Instructor <i>Department of Mathematics, University of British Columbia</i> Math 110/001: Differential Calculus	2018.09–2019.01 2018–2019 Winter Term 1.
Full-time Teaching Assistant <i>Department of Mathematics, University of British Columbia</i> Math 300: Complex Analysis; Math 316: Partial Differential Equations, etc.	2015.09–2021.11

Grants

Active	
1. NSFC, Young Scientists Fund (PI) (30W)	2025.01–
2. NSFC, Original Exploration Program (Co-PI, 130W) (participate)	2025.01–
3. Shanghai Magnolia Talent Plan – Pujiang Project (PI) (30W)	2024.12–
4. Natural Science Foundation of Shanghai, General Project (PI) (20W)	2024.10–
5. Fudan University “AI+” Course Funding (Sub-PI) (20W)	2024.09–
Completed	
1. China Postdoctoral Science Special Fund, In-Station (PI) (18W)	2022.10–2023.11
2. China Postdoctoral Science Fund, General Project (PI) (8W)	2022.10–2023.11
3. International Postdoctoral Exchange Fellowship (PI) (40W)	2021.09–2023.11
4. Shanghai “Super Postdoc” Incentive Plan (PI) (20W)	2021.09–2023.11

Honors & Awards

Fudan University

- Quality Course of Education and Teaching Reform 2026
- Outstanding Postdoctoral Researcher 2025

University of British Columbia

- President's Academic Excellence Initiative PhD Award 2020, 2021
- International Doctoral Fellowship 2017–2021
- International Doctoral Fellowship Tuition Award 2017–2021
- International Tuition Award 2015–2017
- Faculty of Science Graduate Award 2015–2017

The Chinese University of Hong Kong

- First Class Graduate Honor 2015
- Morningside College Master's List 2014–2015
- Science Faculty Dean's List 2014–2015
- Morningside College Exchange Scholarship 2013
- Wei Lun Exchange Scholarships 2013
- Weishan Lake Academic Scholarship 2012, 2013

Relevant Skills

- **Software:** L^AT_EX, MATHEMATICA, MS OFFICE, VISUAL STUDIO.
- **Programming:** C++, C, MATLAB.

Publications

Published / Accepted

1. *On the Spectral Gap of a Square Distance Matrix*, joint with D. Li, D. Shirokoff and B. Wetton, *J. Stat. Phys.*, 2017, 166(3-4), 1029–1035.
2. *Asymptotic Behaviour of Time Stepping Methods for Phase Field Models*, joint with D. Li, K. Promislow and B. Wetton, *J. Sci. Comput.*, 2021, 86(3), 1–34.
3. *On a parabolic Sine-Gordon model*, joint with D. Li, C. Quan and W. Yang, *Numer. Math. Theory Methods Appl.*, 2021, 14(4), 1068–1084.
4. *Non-uniqueness of stationary weak solutions to the surface quasi-geostrophic equations*, joint with H. Kwon and D. Li, *Commun. Math. Phys.*, 2021, 388, 1281–1295.
5. *Global wellposedness for 2D quasilinear wave without Lorentz*, joint with D. Li, J. Xu and D. Zha, *Dyn. Partial Differ. Equ.*, 2022, 19(2), 123–140.
6. *On the equivalence of classical Helmholtz equation and fractional Helmholtz equation with arbitrary order*, joint with D. Li and W. Yang, to appear in *Commun. Contemp. Math.*
7. *Equivalent formulations of the oxygen diffusion problem and other implicit free boundary value problems and implications for numerical approximation*, joint with Z. Fu and B. Wetton, *SIAM J. Appl. Math.*, 2023, 83(1), 52–78.
8. *On the global well-posedness and scattering of the 3D Klein-Gordon-Zakharov system*, joint with J. Xu, *Calc. Var. Partial Differential Equations*, 63(17), 2024.
9. *Localization for general Helmholtz*, joint with D. Li and W. Yang, *J. Differential Equations*, 393, 139–154, 2024.
10. *Energy stable semi-implicit schemes for the 2D Allen-Cahn and fractional Cahn-Hilliard equations*, *IMA J. Numer. Anal.*, 46(2), 758–794, 2026.
11. *Second order energy stable semi-implicit schemes for the 2D Allen-Cahn equation*, *BIT Numer. Math.*, 65(26), 2025.
12. *Stability and convergence of an iterative low-regularity method for the Cahn-Hilliard equation*, joint with Z. Luo and S. Wang, to appear in *J. Differential Equations*.

Preprints

1. *On a Sinc-type MBE model*, joint with D. Li, C. Quan and W. Yang, submitted. ArXiv:2106.16193.
2. *Uniform boundedness of the highest norm for 2D quasilinear wave*, joint with D. Li and J. Xu, submitted. ArXiv:2104.10019.

3. *Energy stability and convergence of Strang splitting method for Cahn-Hilliard equation*, joint with D. Li, in preparation.
4. *Global well-posedness for 2D quasilinear wave equations with non-compactly supported initial data*, joint with D. Li and J. Xu, preprint.
5. *Global well-posedness of a two dimensional wave-Klein-Gordon system with small non-compactly supported data*, submitted. ArXiv:2312.00821.
6. *On semi-implicit schemes for the incompressible Euler equations via the vanishing viscosity limit*, joint with Z. Luo and S. Wang, submitted. ArXiv:2406.12320.
7. *Global well-posedness and uniform-in-time vanishing damping limit for the inviscid Oldroyd-B model*, joint with Z. Luo, Z. Yang and C. Yuan, ArXiv:2410.09340.
8. *A Besov-based integration-by-parts method for the incompressible Navier-Stokes equations*, joint with Z. Luo and S. Wang, ArXiv:2509.23192.
9. *Optimal error estimates of a low-regularity integrator for the SQG equations under minimal regularity*, joint with Z. Luo and S. Wang, submitted.
10. *The vanishing viscosity limit of the surface quasi-geostrophic equations in borderline Besov spaces: Rates and same topology convergence*, joint with Z. Luo and S. Wang, submitted.

Academic Service

I have refereed articles for the following journals:

- **Numerical Analysis, Scientific Computing & Applied Mathematics:**
Mathematics of Computation; IMA Journal of Numerical Analysis; Advances in Computational Mathematics; Applied Mathematics Letters.
- **PDE and Analysis:**
Acta Mathematica Scientia; Advances in Differential Equations; Zeitschrift für Angewandte Mathematik und Mechanik; Communications in Analysis and Mechanics; Fractional Calculus and Applied Analysis; Dynamics of Partial Differential Equations.
- **Mathematical Physics:**
Classical and Quantum Gravity; Physica Scripta.

Mentoring

Ph.D. students

- Haoran Wu (since Fall 2024).
- Weiran Xiong (since Fall 2026).
- Xuewen Lai (since Fall 2026).

Academic Visits and Events

2026.07	Organiser, 11th Shanghai Int. Workshop on Nonlinear Science and Applications <i>Shanghai Society for Nonlinear Science; Guangxi Normal University</i>
2026.04	Speaker, 2026 Lanzhou University Forum on Frontiers of Computational Mathematics <i>Lanzhou University</i> Title: Parameter-stable low-regularity method in fluid models.
2026.03	Speaker, SYSU 5th Anniversary Conf. on PDEs, Dynamical Systems and Computational Mathematics <i>School of Science, Sun Yat-sen University</i> Title: Parameter-stable low-regularity method in fluid models.
2025.11	Speaker, 2025 HKU Workshop on PDEs and Analysis <i>University of Hong Kong</i> Title: Parameter-stable low-regularity method in fluid models.
2025.10	Speaker, 23rd Annual Conference of CSIAM 2025 <i>CSIAM; Xiangtan University; Central South University</i> Title: Parameter-stable low-regularity method in fluid models.

2025.07 **Speaker, 30th Int. Conf. on Difference Equations and Applications (ICDEA 2025)**
Int. Soc. Difference Equations; Guangzhou University
 Title: Analytic and numerical results of PDE models in material sciences and biology.

2025.07 **Co-organiser, 2nd Future-oriented Complex Systems Science (FOCSS) Workshop**
IICS, Fudan University

2025.06 **Participant, Advances in Mathematical Fluid Dynamics**
Peking University

2025.06 **Speaker, Workshop on Differential Equations and Dynamical Systems**
SUSTech
 Title: Parameter stability in fluid and related models.

2025.05 **Speaker, Workshop on Theory, Scientific Computing and Optimization of PDEs**
Chongqing Normal University
 Title: Parameter stability in fluid and related models.

2024.12 **Co-organiser, 1st Workshop on Differential Equations and Nonlinear Analysis**
Fudan University

2024.06 **Speaker, 4th Int. Symposium on Modelling, Analysis and Applications in Biomathematics**
Harbin Engineering University
 Title: Analytical and numerical results of PDE models from material sciences and biology.

2024.06 **Speaker, Workshop on Nonlinear Analysis**
Wuhan University of Technology
 Title: Analytical and numerical results of PDE models from material sciences and biology.

2024.01 **Participant, Sanya Bo International Frontier Forum**
Tsinghua Sanya International Mathematics Forum

2023.11 **Co-organiser and Speaker, 1st FOCSS Workshop**
IICS, Fudan University

2023.11 **Speaker, Seminar Talk**
Institute of Mathematical Sciences, ShanghaiTech University
 Title: Analytical and numerical results of some phase field and free boundary models.

2023.07–2023.08 **Academic Visitor**
South China University of Technology

2022.12 **Speaker, Online Workshop on Analysis and Computation of PDEs**
SUSTech & Beijing Normal University (Zhuhai)
 Title: Trigonometric type models in the recent study of phase field problems.

2021.02–2021.04 **Academic Visitor**
SUSTech International Center for Applied Mathematics

2019.08 **Speaker, Workshop on Analysis and PDE**
Tianjin Center for Applied Mathematics (TCAM)
 Title: Fully implicit time stepping for Allen-Cahn dynamics.

2019.06–2019.08 **Academic Visitor**
Tianjin University
 Invited by Prof. Yifei Wu.

2019.05 **Speaker, SIAM Conf. on Applications of Dynamical Systems (Minisymposium)**
SIAM, Snowbird, UT, USA
 Title: Fully implicit time stepping for Allen-Cahn dynamics.

2018.04 **Academic Visitor**
Michigan State University
 Invited by Prof. Keith Promislow.

2016.08 **Participant, PIMS Graduate Mathematical Modelling in Industry Workshop**
Pacific Institute for the Mathematical Sciences, Vancouver, BC, Canada